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Mesoporous boron carbon nitride/graphene modified separators as efficient polysulfides barrier for highly stable lithium-sulfur batteries

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Abstract

Lithium-sulfur batteries (LSBs) have drawn intense attention due to their high theoretical specific energy and low material cost. However, shuttle effect of the lithium polysulfides (LiPSs) dissolved in the electrolyte results in severe capacity fading and self-discharge. Herein, a multifunctional modifier composed of mesoporous boron carbon nitride and conductive graphene (BCN/G) has been developed to modify separator as an efficient barrier for LiPSs in LSBs. Benefiting from such a modified separator, LSBs exhibit improved cycling capacity and high rate capacities. The excellent electrochemical performances are mainly attributed to the strong chemical interactions between LiPSs and conductive BCN/G modifier, which have been confirmed by X-ray photoelectron spectroscopy analyses. Such a design of separator opens a new approach for high-performance LSBs.

Keywords: Lithium-sulfur batteries; Shuttle effect; Boron carbon nitride; Graphene; Separator

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1. Introduction

Rechargeable lithium-sulfur batteries (LSBs) have attracted much attention due to their remarkable theoretical energy density (2567 Wh kg^{-1}), low cost, environmental compatibility, and natural abundance of sulfur on the earth [1-4]. Although LSBs have shown great application potential, it still faces serious challenges that hinder their uses in practice. The so-called shuttle effect in LSBs is a major issue [5-7]. Upon cycling, the formed lithium polysulfides (LiPSs) shuttle through the separator to lithium anode, resulting in loss of the active substances, reduced Coulombic efficiency, poor cycle performance, and severe self-discharge [8-16]. In addition, the insulating nature of the elemental sulfur leads to sluggish reaction kinetics and poor cycling performance of the cells [17-22].

A great deal of researches has been conducted to solve the mentioned problems. A commonly used and effective method is to encapsulate sulfur into host material such as carbon-based materials, metal materials, and metal-organic frameworks [23-25]. Previous researches showed that encapsulating sulfur into various host materials can effectively inhibit shuttle effect of the soluble LiPSs and significantly improve cycle performance of the batteries [26-28]. However, preparation of these host materials often needs high-cost raw materials and complicated processes, which are not suitable for large-scale production [29, 30]. It is necessary to develop a low-cost material to facilitate the adsorption of LiPSs, and the preparation method is relatively simple.

Modifying the separator can effectively inhibit the shuttle effect. Porous carbon, conductive polymers, graphene, and carbon nanotubes (CNTs) have been coated on the separators to improve the electrochemical performance of LSBs [31-35]. Zhou *et al.* proposed to integrate sulfur and graphene onto separator, a significant increase in the conductivity of the cathode was obtained due

to the composited graphene accelerating electron transport [36]. Huang *et al.* synthesized conductive porous CNTs/graphene oxide hybrid interlayer for LSBs [37]. They found that such a carbon-based material can not only suppress the shuttling of LiPSs, but also increase the electrical conductivity of the sulfur cathode. However, carbon-based materials are mostly physical barriers to the soluble LiPSs. Recently, materials containing chemisorption to LiPSs have been introduced to modify the separators and positive electrodes for restraining the shuttle of LiPSs. This kind of polar materials can chemically bond with LiPSs, resulting in the suppressed shuttle effect and better electrochemical performance [38]. Mi *et al.* synthesized nitrogen and oxygen co-doped nanoporous carbonaceous materials to suppress the shuttle effect in LSBs [39]. The polarized carbon host possess abundance of active sites on its surface to ensure the chemical confinement of LiPSs. Pang *et al.* developed an ordered nanoporous carbon co-doped with nitrogen and sulfur to chemisorb the soluble LiPSs [40]. Song *et al.* reported a preparation of the mesoporous nitrogen-doped carbon@sulfur nanocomposite as a cathode material for LSBs, and demonstrated that nitrogen doping can effectively reduce LiPSs with chemisorption [41]. Babu *et al.* produced a BN-modified separator as a barrier for LiPSs [42], they found that BN can chemically bond with LiPSs to suppress the shuttle effect. Fan *et al.* applied graphene and functional boron nitride in the sulfur electrode to block the LiPSs [43]. Other than the sulfur redox reaction is carried out in a variety of interface, liquid-solid interconversion and solid-solid interconversion and the like [44], which includes the conversion of the soluble long-chain LiPSs into Li_2S precipitates. In order to speed up the conversion of lithium polysulfide, the researchers added mediator to promote redox kinetics and increase reaction efficiency [45,46].

In this work, we combined mesoporous boron carbon nitride (BCN) with graphene nanosheet as an efficient modifier for separator in LSBs. The modified separator can be easily prepared by vacuum

filtration. Since BCN and graphene have strong chemical adsorption and physical adsorption to the soluble LiPSs, respectively, and graphene has excellent electrical conductivity [42, 43], a multifunctional BCN/graphene (BCN/G) modifier was obtained with good electrical conductivity and strong chemical adsorptions to improve the performance of LSBs. BCN microbelts intercalate in the graphene to form a stable structure, which can be firmly attached to the pristine polypropylene separators. The modified separator can effectively obstruct the shuttle effect of LiPSs and decrease the polarization effect of the sulfur cathode. Consequently, LSBs with the BCN/G modified separator exhibit excellent electrochemical performance. It is worth mentioning that the preparation of the BCN microbelts is simple, and BCN is a metal-free material, it is lighter than other modifiers containing metal elements, providing a solution for reducing the weight of the battery system.

2. Experimental Section

2.1. Polysulfide solution preparation

The dark brown Li_2S_6 solution was prepared by mixing lithium sulfide and sulfur power with a molar ratio of 5:8 into a DOL/DME (v/v = 1:1) solution, and stirred at 60 °C for 24 h. The whole process was carried out in an argon-filled glovebox ($\text{H}_2\text{O} < 0.5$ ppm, $\text{O}_2 < 0.5$ ppm, MBraun, Unilab).

2.2 Sulfur cathode preparation

Sulfur cathodes were prepared by a slurry coating method. The slurry is composed of 70 wt% sulfur, 20 wt% Ketchen black, and 10wt% polyvinylideneuoride (PVDF) in Nmethyl-2-pyrrolidone (NMP), which was coated on carbon-coated aluminum foil and dried at 60 °C overnight in a vacuum oven. The mass loading of sulfur in the resultant cathode is in the range of 1.3-6.1 mg cm^{-2} .

2.3. Synthesis of porous BCN microbelts

Boric acid and melamine (in 2:1 molar ratio) were dissolved in deionized (DI) water at 90 °C, stirring until the solution is completely dissolved and transparent. The solution was again heated to 300 °C to evaporate water. The dried precursor was placed in a quartz tube furnace, and heated to 1000 °C with a rate of 5 °C min⁻¹ for 3 h. The obtained BCN was ball milled overnight and then screened through a 200 mesh sieve.

2.4. Synthesis of BCN/G modifier

The BCN/G coated separator was obtained by a vacuum filtration method. 5 mg BCN and 5 mg graphene were dispersed in 100 ml NMP by sonication for 1h to make a homogeneous suspension. Then the suspension was filtered through a polypropylene membrane (Celgard 2400) to form a smooth BCN/G modified separator with BCN/G loading of 0.47 mg cm⁻². The modified separator was punched into a round shape with a diameter of 19 mm. The contrast sample is a pure graphene modified separator which was fabricated using the same process with 0.49 mg cm⁻² graphene loading in total. The pure BCN modified separator was made too, which has the same areal mass loading.

2.5. Characterization method

The morphologies and the corresponding elemental mappings of BCN and graphene were observed by scanning electron microscopy (SEM) equipped with an energy dispersive X-ray spectroscopy (EDS) and high-resolution transmission electron microscopy (TEM, FEI Tecnai G2 F30) using an accelerating voltage of 300 kV. The crystal structure of the materials was characterized by X-ray powder diffraction (XRD, Malvern Panalytical Empyrean with Cu K α radiation). Brunauer-Emmett-Teller (BET) surface areas were deduced based on the nitrogen adsorption-desorption isotherms recorded by a micromeritics ASAP 2020 accelerated surface area and porosimetry system in N₂ atmosphere. X-ray photoelectron spectroscopy (XPS) was measured

through a Kratos axis ultra DLD instrument with Al K α probe beam. Ultraviolet-visible (UV-Vis) transmission test was performed on a Shimadzu UV-3600 spectrophotometer.

2.6. Electrochemical measurements

Electrochemical performance was tested by using CR-2032 cells which were assembled in a high-purity argon-filled glovebox ($\text{H}_2\text{O} < 0.5$ ppm, $\text{O}_2 < 0.5$ ppm, MBraun, Unilab) with lithium foil as the reference/counter electrode. The electrolyte used was 1 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) dissolved in a mixture of 1,2-dimethoxyethane (DMC) and 1,3-dioxolane (DOL) at a volume ratio of 1:1 containing 2.0 wt% LiNO_3 . BCN/G modified Celgard 2400 polypropylene membrane was used as the separator. For comparison, Celgard 2400 polypropylene membranes modified respectively by BCN and graphene, and Celgard 2400 polypropylene membrane only were used as the separators. Galvanostatic discharge-charge tests were performed in a voltage window of 1.8-2.7 V using a multichannel battery tester (Neware BTS-610). The electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV) were carried out at room temperature by using an electrochemical workstation (CHI 660C).

3. Results and discussion

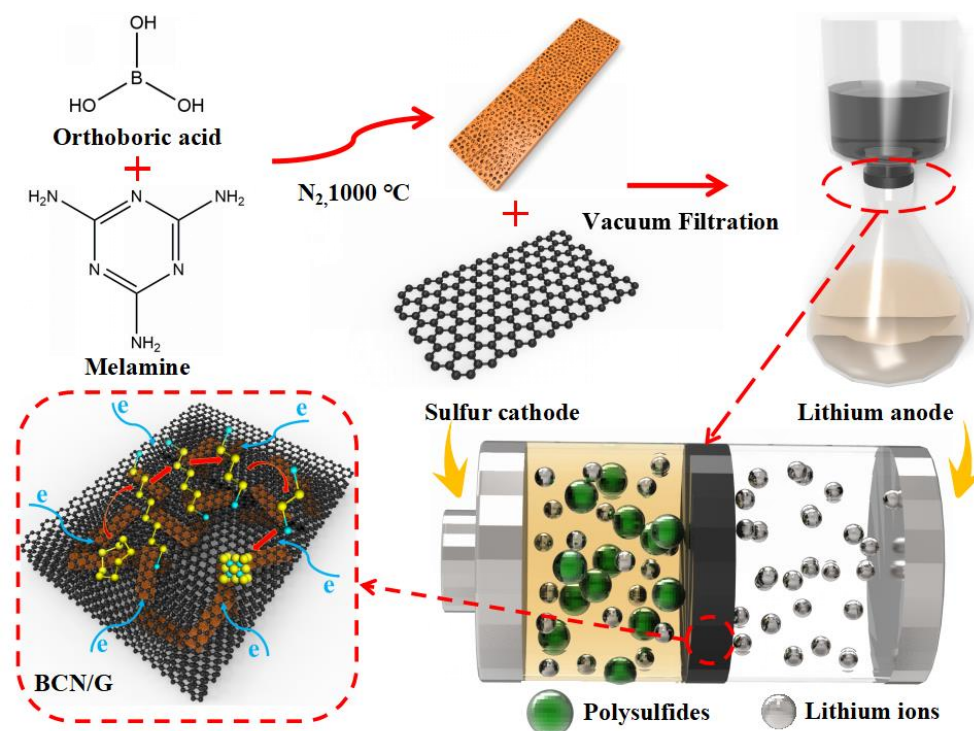


Fig. 1 Schematic illustration of synthesizing the BCN/G modifier and electrode configuration of the LSB with a BCN/G modified separator.

The schematic diagram of the experimental process for synthesizing BCN/G modifier is shown in Fig 1. Boron comes from boric acid while carbon and nitrogen are derived from melamine. The obtained BCN and commercial graphene were mixed together and vacuum filtered onto polypropylene separator. The LiPSs produced during the discharge can easily diffuse from the cathode to anode through the separator by a concentration gradient force. Unlike traditional separators, BCN/G modified separator has strong chemical adsorption to LiPSs which can be effectively blocked from shuttling. On the other hand, the addition of graphene sheets can greatly improve the conductivity, and has a certain physical adsorption effect on LiPSs through their two-dimensional layered structure.

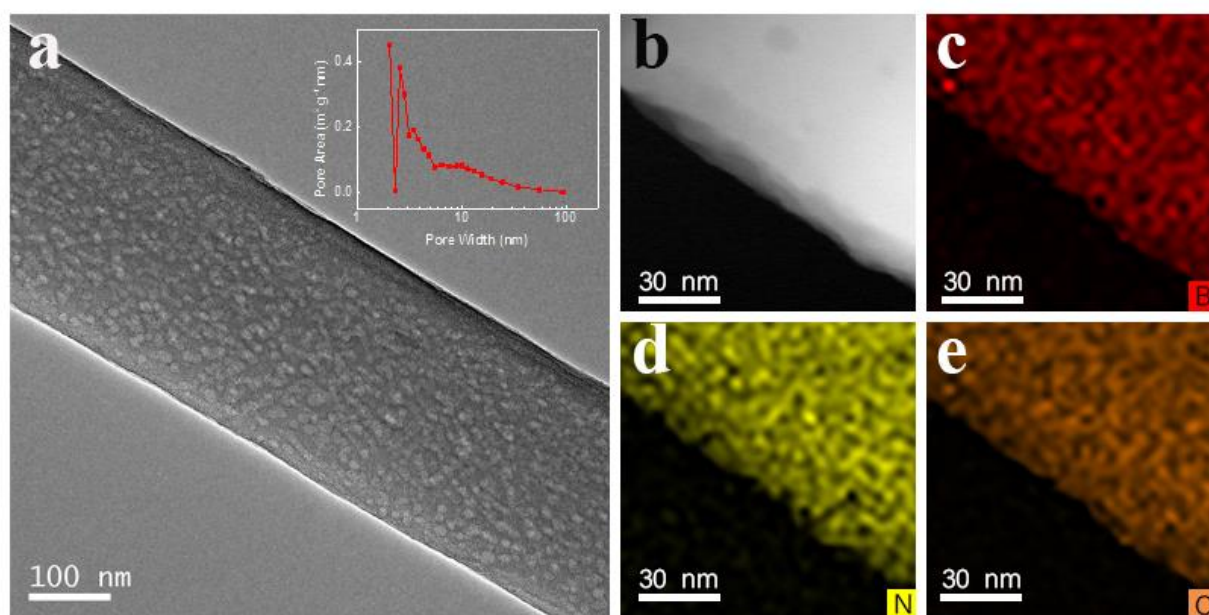


Fig. 2 (a) TEM image and (b-e) STEM image and the corresponding elemental mappings of the synthesized BCN microbelts. The inset in (a) shows the pore size distribution.

Numerous small holes can be clearly seen from the TEM image of the synthesized BCN microbelts as shown in Fig. 2a, and the BET measurement shows that their structure is mesopores. The specific surface area of the BCN microbelts is $36.83 \text{ m}^2 \text{ g}^{-1}$, and the average pore width is 18.4 nm. In addition, the micropore volume is estimated to be $0.003990 \text{ cm}^3 \text{ g}^{-1}$. The pore volume of pores with diameter less than 352.2 nm is $0.169808 \text{ cm}^3 \text{ g}^{-1}$ (Fig. S1). The mesopores can act as traps to capture the soluble LiPSs, increasing the reaction areas and improving the reaction efficiency between BCN and LiPSs. The synthesized BCN was further characterized by TEM elemental mappings of carbon, boron, and nitrogen (Fig. 2b-e). All of the elements were found to be evenly distributed throughout the sample. BCN microbelts were synthesized by a simple template-free two-step reaction including precursor solidification and high-temperature pyrolysis, their surface is rich in pores that can act as active sites for polysulfide reactions. Special B-N bond polarity can effectively bond with polysulfides. From XRD patterns shown in Fig. S2, we can

clearly detect similar peaks at 25.4° and 42.5° , corresponding to (002) and (100) planes of BCN, respectively. The peaks are wider for BCN, suggesting the smaller particle size and poor crystallization. The calculated lattice parameter of the (002) planes of BCN is 0.34 nm based on the XRD pattern, which is consistent with the fringe spacing in the high-resolution TEM image shown in Fig. S3.

The morphology of the BCN/G modified separator was observed by SEM. It can be seen that the BCN microbelts which are marked with red circles shown in Fig. S4a, are uniformly distributed in the graphene sheets. The BCN/G modifier was vacuum filtered on the pristine separator to form a barrier layer with a thickness of 40 μm (Fig. S4b). As shown in Fig. S4c, the commercial separator displays abundant pores of around 100 nm in size. The BCN/G modifier was further characterized by SEM elemental mappings of carbon, boron, and nitrogen (Fig. S4d-g), all of these elements were found to be evenly distributed throughout the sample.

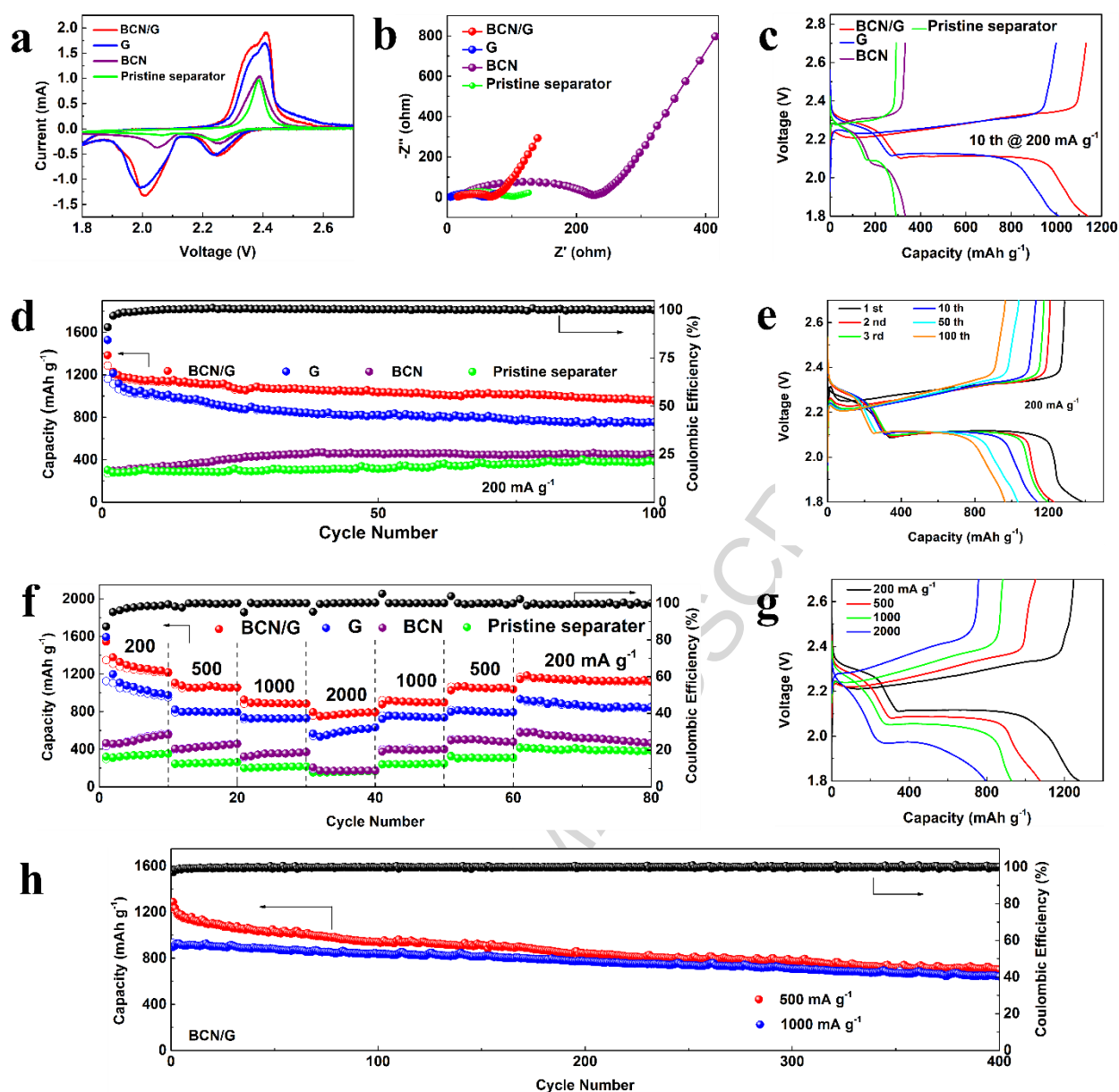


Fig. 3 (a) CV curves and (b) EIS spectra of the cells with various modified separators and pristine separator. (c) Discharge-charge curves of the cells in the 10th cycle at 200 mA g⁻¹. (d) Cycle performance of the cells at a current density of 200 mA g⁻¹. (e) Discharge-charge curves of the cell with the BCN/G modified separator at 200 mA g⁻¹. (f) Rate performance of the cells within a potential window of 1.8–2.7 V. (g) Discharge-charge curves of the cell with the BCN/G modified separator at 200, 500, 1000, and 2000 mA g⁻¹. (h) Long-term cycling stability of the cell with the BCN/G modified separator at 500 and 1000 mA g⁻¹.

In order to evaluate the effect of the BCN/G modified separator on the electrochemical performance of LSBs, the coin cells were assembled using sulfur electrode as the cathode and metallic lithium as the anode. The modified Celgard 2400 polypropylene membranes with various modifiers were respectively used as the separators. Fig. 3a compares CV curves of the cells respectively with the BCN/G, graphene, BCN modified separators and pristine separator. The two cathode peaks correspond to the formation of Li_2S_x ($x=4\sim 8$) and $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. The strong reduction peaks of the LSB with the BCN/G or graphene modified separator are located at 2.01 V and 1.98 V, respectively. For the LSB with the pristine separator or BCN modified separator, only one anode peak can be observed at 2.40 V. For the LSB with the BCN/G modified separator, it can be seen that there are two anode peaks at 2.37 and 2.40 V, which correspond to the lithium sulfide being oxidized to higher-order LiPSs and further oxidized to S_8 , respectively. The higher intensity and larger area of the oxidation and reduction peaks suggest that the electrochemical reactions are more sufficient for the LSB with the BCN/G modified separator. By comparing the CV curves of the four cells, it can be concluded that the LSB with the BCN/G modified separator has the maximum capacity.

EIS measurements were carried out to understand the electrode kinetics and interface characteristics of the LSBs. Fig. 3b compares EIS spectra of the LSBs with various modified separators and pristine separator, the fitted data are shown in Table S1. The charge transfer resistance (R_{ct}) of the cell with the BCN/G modified separator is as low as 15.5 Ω , this is most likely due to the higher conductivity of graphene nanosheets and strong affinity of BCN microbelts for LiPSs. The discharge-charge curves of the LSBs with various separators were tested at a current density of 200 mA g^{-1} , and the 10th discharge-charge curves are shown in Fig. 3c. The specific capacity of the cell with the BCN/G modified separator is the highest as reflected by the CV curves shown in Fig 3a. BCN has a larger voltage hysteresis, it can be significantly reduced after mixing with graphene

sheets. Commercial graphene was used in the experiment, and graphene was simply characterized (Fig. S5). As shown in Fig. 3d, a capacity of 965 mAh g^{-1} is remained for the LSB with the BCN/G modified separator after 100 cycles at 200 mA g^{-1} , and the cycling stability is distinctly superior to the other cells. The LSB with the pristine separator shows low capacities and poor cycling stability. The electrochemical performance of the cell with the BCN modified separator is degraded due to the poor conductivity of the BCN microbelts. The mass loading of sulfur in these cells is higher than 1.3 mg cm^{-2} . The different proportions of BCN and graphene in the modifiers were designed to determine the optimum composition (Fig. S6), and the BCN/G modifiers with different thickness were prepared to optimize the thickness (Fig. S7). The optimal composition ratio of BCN to graphene is 1:1 by mass ratio, and the optimum thickness of the BCN/G modifier was obtained after filtering 10 ml of the BCN/G solution onto the polypropylene membrane (Fig. S8). The discharge-charge curves of the LSB with the BCN/G modified separator cycled at 200 mA g^{-1} (Fig. 3e) display two typical discharge plateaus at 2.3 and 2.1 V, which correspond to the two processes of S_8 being converted to Li_2S_6 and then to Li_2S_4 . Then Li_2S_4 is converted to Li_2S_2 and Li_2S , respectively. Two platforms appear in the charge process, indicating that lithium sulfides are oxidized to LiPSs and S_8 . The results here are similar to those obtained by CV measurement. There was no significant change in the voltage hysteresis after 100 cycles. It can be concluded that the LSB with the BCN/G modified separator has excellent stability. The rate capability of the cells with various modified separators and pristine separator were evaluated as shown in Fig. 3f and S6-S7. The highest capacity of 794 mAh g^{-1} at 2000 mA g^{-1} was obtained for the LSB with the BCN/G modified separator. It can be seen that the cell shows less polarization, in contrast the chemical reactions of the other cells are not sufficient and their polarizations are serious. Fig. 3g displays the discharge-charge curves of the cell with the BCN/G modified separator at various current densities.

The long cycling performance of the cell was investigated at larger currents. As shown in Fig. 3h, the higher capacities of 701 and 645 mAh g⁻¹ are available after 400 cycles at 500 and 1000 mA g⁻¹, respectively.

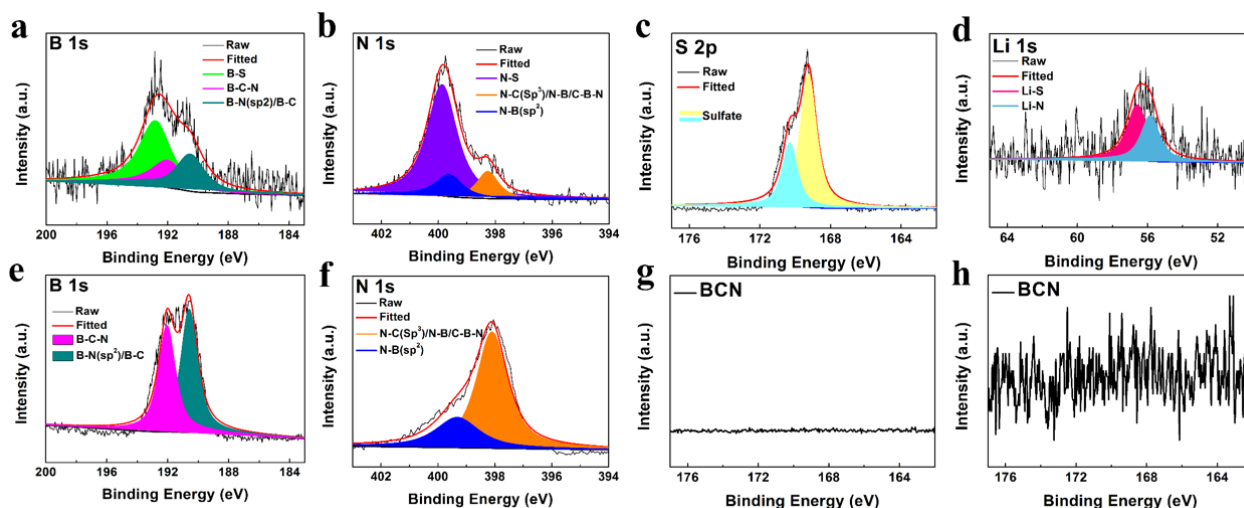


Fig. 4 High-resolution XPS spectra of (a) B 1s, (b) N 1s, (c) S 2p, and (d) Li 1s in the cycled BCN modified separator. High-resolution XPS spectra of (e) B 1s, (f) N 1s, (g) S 2p, and (h) Li 1s in the as-prepared BCN modified separator.

To explore the action mechanism between BCN and LiPSs and explain the improvement of the electrochemical performance observed in the LSBs with the BCN/G modified separator, we performed XPS characterization on the BCN modified separator before and after 5 discharge-charge cycles. As shown in Fig. 4, XPS analysis of the cycled BCN modified separator shows sulfate, lithium sulfide and lithium nitrides signals. However, these chemical compounds are not present in the as-prepared BCN modified separator. Therefore, the LiPSs generated during discharge-charge cycles have been trapped on the functional separator. As shown in Fig. 4a and e, the B 1s core energy spectrum of the as-prepared BCN modified separator shows two peaks at 192.03 and 190.55 eV, which can be attributed to the B-C-N and B-N(sp²)/B-C bonds, respectively. After the discharge-charge cycles, a strong peak corresponding to B-S bonds appears at 192.83 eV, indicating

a chemical reaction between LiPSs and B in BCN. In order to prove BCN having a chemisorption effect on LiPSs, N element was also analyzed by XPS. As shown in Fig. 4b and f, the as-prepared BCN modified separator has two XPS peaks at 399.35 and 398.10 eV, which correspond to the N-B(sp²) and N-C(sp³)/N-B/C-B-N bonds, respectively. After the discharge-charge cycles, a strong peak that belongs to the N-S bonds appears at 399.89 eV. In conclusion, the XPS analysis showed that BCN has strong chemisorption for LiPSs.

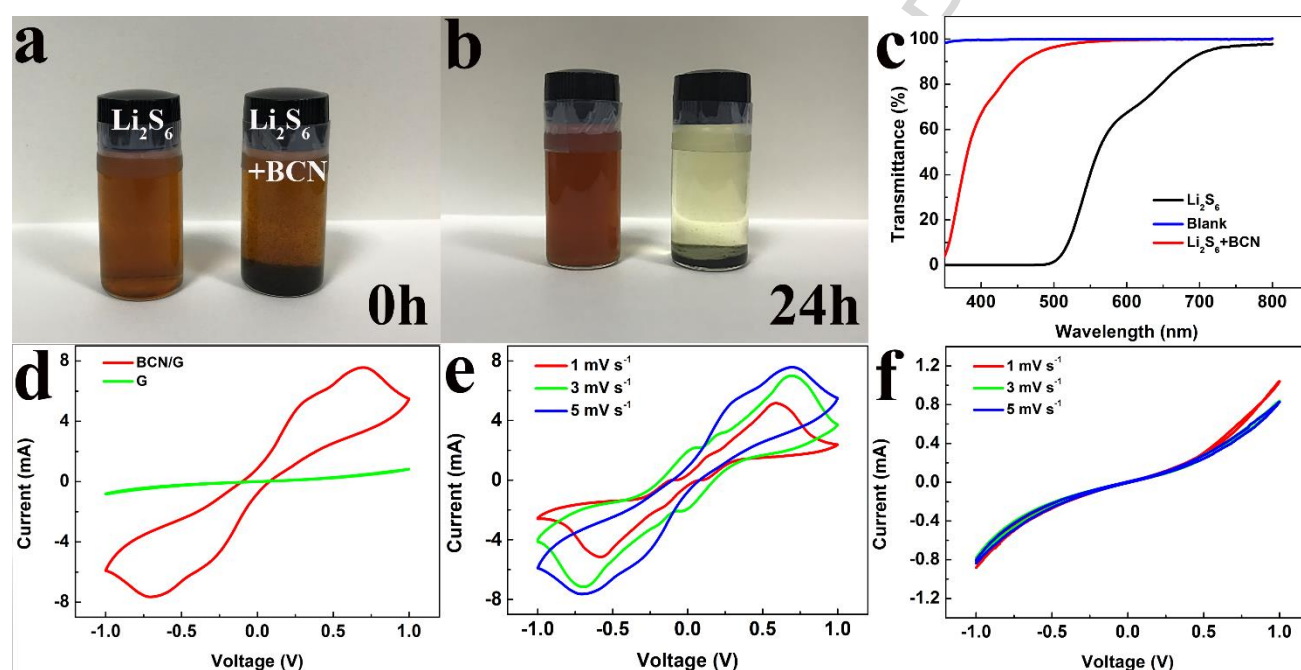


Fig. 5 (a, b) Optical photographs of the Li₂S₆-containing solutions with/without the BCN powder after 0 h and 24 h standing. (c) UV-vis spectra of the Li₂S₆-containing solutions with/without the BCN powder after 24 h standing. (d) CV curves of the symmetric dummy cells employing BCN/G and pristine G electrodes at a rapid scan rate of 5 mV s⁻¹. (e, f) CV curves of BCN/G and G at different scan rates.

In order to better demonstrate the adsorption of BCN for LiPSs, we added the prepared BCN to a solution of 0.02 M Li₂S₆, and a pure Li₂S₆ solution was prepared for comparison. As shown in Fig. 5a and b, the color of the Li₂S₆ solution with the BCN adding changes significantly, while the pure

Li_2S_6 solution is almost unchanged after 24 h standing. The Li_2S_6 solution with the BCN adding is almost transparent by visual inspection after 24 h standing, indicating that LiPSs have been adsorbed by BCN and precipitated to the bottom of the bottle. The adsorption of BCN for LiPSs was also analyzed by measuring UV-vis spectroscopy. In general, various polysulfide chain mixtures (S_8^{2-} , S_6^{2-} , S_4^{2-} and $2\text{S}_6^{2-} \rightarrow \text{S}_8^{2-} + \text{S}_4^{2-}$) can be dissolved in DOL:DME solvent, and the LiPSs solution exhibits an ultraviolet absorption in the wavelength range of 400-500 nm, which is affected by the concentration of LiPSs in the solvent. As shown in Fig. 5c, the substrate tested is DOL:DME solvent, its transmittance is almost 100%. In contrast, the transmittance of the dissolved LiPSs solution with a higher concentration is small in the wavelength range of 400-500 nm. However, for the same concentration of LiPSs solution with the BCN adding after 24 h standing, its average transmittance is 81.85% in the same wavelength range, indicating that the concentration of LiPSs in the solution is lower as a result of BCN adsorption.

In order to further gain a understanding of the strong chemical interactions and possible electrocatalysis between BCN and LiPSs, the performance of redox reactions was investigated by CV measurement using a symmetrical dummy cell with Li_2S_6 containing electrolyte (Fig. 5d-f). The current density of the BCN/G electrode was much higher than that of the G electrode, indicating that the redox kinetics reaction between LiPSs was significantly enhanced after the addition of BCN. The catalytic properties of BCN were further elaborated by CV curves measured under different scanning rates. The redox current density increases linearly with increasing the scanning rate, demonstrating that it has a strong diffusion-controlled characteristic for the electrochemical conversion. In general, the diffusion of the active phases lags far behind their depletion of redox at rapid scan rates, the enhanced redox reaction of the soluble LiPSs is mainly attributed to the strong chemisorptions and electrocatalysis of BCN that triggers the rapid conversion of LiPSs.

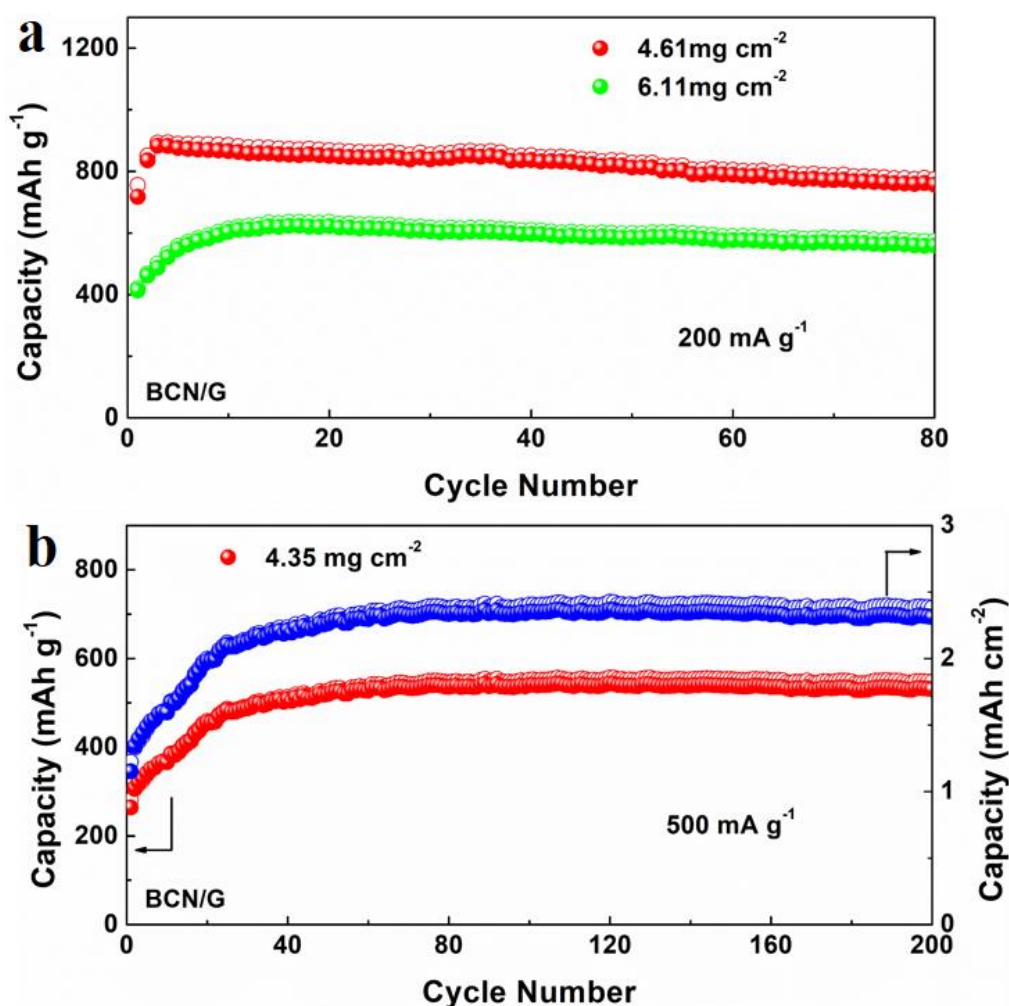


Fig. 6 Cycle performances of the LSBs with the BCN/G modified separator and different active material loadings at current densities of (a) 200 and (b) 500 mA g⁻¹.

High loaded sulfur cathode is a key for the high-energy-density LSBs in practical applications. In order to evaluate the electrochemical performance of the LSBs with the BCN/G modified separator under high loading of sulfur, sulfur cathodes with higher active material loading (4.6-6.1 mg cm⁻²) were prepared and performed a series of electrochemical measurements. As shown in Fig. 6a, the cathode with an active material loading of 4.6 mg cm⁻² in the LBS exhibits a reversible capacity of 774.00 mAh g⁻¹ after 80 cycles at 200 mA g⁻¹. At a loading of 6.11 mg cm⁻², the battery still exhibits good stability, a capacity of 573.60 mAh g⁻¹ is remained after 80 cycles at 200 mA g⁻¹. When cycled at 500 mA g⁻¹ for 200 times, the cathode with an areal mass loading of 4.35 mg cm⁻² exhibits a higher capacity of 546.50 mAh g⁻¹ (Fig. 6b), corresponding to an areal capacity of 2.38 mAh cm⁻².

4. Conclusion

In conclusion, we have prepared a lightweight and multifunctional separator composed of mesoporous BCN microbelts and conductive graphene sheets coated on polypropylene membrane for LSBs using a simple and low-cost vacuum filtration method. The representative LSB with such a separator shows a high residual capacity of 837.74 mAh g⁻¹ after 100 cycles at 1000 mA g⁻¹. Due to the strong chemical adsorption, the prepared BCN/G modifier can efficiently adsorb the soluble LiPSs. High conductivity and porosity of the modifier are in favor of the rapid transfer of ions/electrons and electrolyte penetration. The N-functional groups on BCN play an important role in preventing LiPSs from shuttling. The strategy of developing the BCN/G modified separator also provides a solution for controlling ion transmission for other energy storage systems such as lithium-ion batteries and supercapacitors.

Acknowledgments

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Highlights

- A MULTIFUNCTIONAL BCN/G MODIFIED SEPARATOR WAS DESIGNED AS EFFICIENT BARRIER FOR LiPSS.
- The BCN/G based separator act as a chemical trapper for LiPSS and an accelerator for the redox kinetics of sulfur species
- THE CHEMICAL INTERACTIONS BETWEEN BCN AND LiPSS ARE CONFIRMED BY XPS ANALYSES.
- LSBs WITH BCN/G MODIFIED SEPARATORS EXHIBIT EXCELLENT ELECTROCHEMICAL PERFORMANCE.